

Intro. to Probability:

Recall 2 Definitions:

Def. (Experiment): An activity or process whose outcome is: i) of interest, ii) subject to uncertainty.

Def. (Sample Space): Given an experiment, the sample space S is the set of all possible outcomes.

We now add a third that is closely related to the sample space.

Def. (Event): An event is any subset of the sample space S .

If an event contains exactly one item, it is called a simple event. Otherwise, it is a compound event.

Ex: Events where $S = \{1, 2, 3, 4, 5, 6\}$

- $A = \{1\}$, "the event that the die reads 1"
- $A = \{2, 4, 6\}$, "the event that the die is even"

Ex: Rolling 2 dice. $S = \{(1, 1), (1, 2), (1, 3), \dots, (6, 6)\}$

- $A = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$
"event that the two dice sum to 7"
- $A = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6)\}$
"event that the first die equals 1"

Our goal is to be able to assign to all events a "measure" that tells us how likely that event is to occur.

Such a measure is called **probability**. The probability of event A , where A is some event, will be denoted by **$IP(A)$** .

Axioms of Probability: Consider an experiment with sample space S . The probability measure $IP()$ must satisfy:

1) For any event A , $IP(A) \geq 0$ [Non-negativity]

2) $IP(S) = 1$ [Unit Measure]

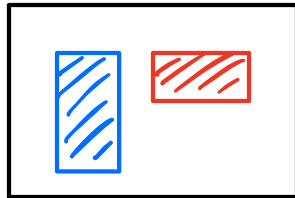
3) If $A_1, A_2, \dots$ is a countably infinite collection of mutually exclusive events, then $IP(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} IP(A_i)$
[Countable Additivity]

Note: Intuition - we want probability to behave like other measures, e.g. length, area, volume, etc. Only its 'measure' is likelihood.

1) Measures can't be negative, e.g. something can't have area of $-1m^2$.

2) Easiest to see in an example - a coin, $S = \{H, T\}$. It will always be true that the coin is either H or T - otherwise $\{H, T\}$ would not be the true sample space. By convention, we associate "always" with probability measure of 1.

3) Consider 2 rugs on the floor, that don't overlap with each other. The area covered by either one or the other (i.e. area of the union) is just equal to the sum of each's area. Note that this requires that the two don't overlap (i.e. disjoint).



Ex.: Probabilities in Some Experiments

- Tossing a "fair" coin. $S = \{H, T\}$

$$\leadsto P(\{H\}) = 1/2, \quad P(\{T\}) = 1/2$$

Exercise: Show $P(S) = 1$

- Rolling a "fair" die. $S = \{1, 2, 3, 4, 5, 6\}$

$$\leadsto P(\{1\}) = P(\{2\}) = \dots = P(\{6\}) = 1/6$$

Exercise: Show $P(S) = 1$

Note: Usually we are given (or specify) the probability of each simple event. Since any 2 simple events are always disjoint, Axiom #3 can tell us the probability of compound events.

Properties of Probability: Let S be the sample space of some experiment, and A, B be two events. If $P(\cdot)$ is a probability measure (i.e. satisfies Axioms 1-3), then:

i) $P(\emptyset) = 0$

ii) If $A \subset B$, then $P(A) \leq P(B)$ [Monotonicity]

iii) $P(A) \leq 1$

iv) $P(A^c) = 1 - P(A)$

v) $P(A) = P(A \cap B) + P(A \cap B^c)$ [Law of Total Probability]

vi) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

vii) $P(A \cup B) \leq P(A) + P(B)$ [Union Bound]

Proof: (Some but not all)

i) Let $A_1, A_2, \dots, A_n = \emptyset$. Notice that $\bigcup_{i=1}^n A_i = \emptyset$. But by Axiom 3,

$$\begin{array}{ccc} P(\bigcup_{i=1}^n A_i) & = & \sum_{i=1}^n P(A_i) \\ \text{"} & & \text{"} \\ P(\emptyset) & & P(\emptyset) \end{array}$$

Hence: $P(\emptyset) = n \cdot P(\emptyset) \leadsto P(\emptyset) = 0$ ~~$\square$~~

ii) Let $A_1 = A$, $A_2 = A^c \cap B$. Notice that $A_1 \cup A_2 = B$.

(Why? Because $A_1 = A \cap B = A$ due to $A \subset B$).

Since A_1 and A_2 are disjoint, Axiom 3 gives:

$$P(A, \cup A_2) = P(A_1) + P(A_2)$$

$$P(B) = P(A) + P(A^c \cap B).$$

Conclude with $P(A^c \cap B) \geq 0$ by Axiom 1. $\square$

iii) Use Property ii, with $B=S$. This gives:

$$P(A) \leq P(S)$$

Conclude with $P(S)=1$ by Axiom 2. $\square$

Note: The Law of Total Probability and Union Bound generalize to more than 2 events as follows:

- If $B_1, B_2, \dots, B_n$ is a collection of mutually exclusive and collectively exhaustive events, then $P(A) = \sum_{i=1}^n P(A \cap B_i)$.

- If $A_1, \dots, A_n$ are any events, then $P(\bigcup_{i=1}^n A_i) \leq \sum_{i=1}^n P(A_i)$.

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In the vast majority of cases, we are given both the sample space S , and the probability of each simple event.

From this information, we may then want to compute the probability of a compound event.

Ex: Rolling a "fair" 6-sided die

$$S = \{1, 2, 3, 4, 5, 6\}, \quad P(\{1\}) = \dots = P(\{6\}) = 1/6$$

$$A = \text{die roll is even} = \{2, 4, 6\}$$

$$\begin{aligned}
 IP(\{2, 4, 6\}) &= IP(\{2\} \cup \{4\} \cup \{6\}) \\
 &= IP(\{2\}) + IP(\{4\}) + IP(\{6\}) \quad \text{Axiom 3} \\
 &= \underline{3 \cdot \frac{1}{6}} = \frac{1}{2}
 \end{aligned}$$

Notice that there are 3 items in event A , and its probability was 3x the probability of any simple event.

Why? Equally Likely Outcomes

Def. (Equally Likely Outcomes): An experiment with N possible outcomes (i.e. $|S| = N$) is said to have equally likely outcomes if for any simple event A , $IP(A) = \frac{1}{N}$.

This is an extremely nice property due to the following.

Prop.: If an experiment has equally likely outcomes, then for any event A , $IP(A) = \frac{|A|}{|S|} = \frac{\# \text{ outcomes in } A}{\# \text{ outcomes in } S}$.

Proof: Use Axiom 3 and eq. lik. outcomes. $\square$

Takeaway: With equally likely outcomes, computing probabilities amounts to counting! (# of outcomes in S , and in some event)

Counting Techniques:

Sometimes an Experiment consists of several "smaller" sub-experiments. E.g. Flipping 2 coins. It's useful to relate the sizes of the different sample spaces.

Prop.: If we have k different "sub-experiments", whose number of possible outcomes are $N_1, N_2, \dots, N_k$, then the overall experiment (consisting of all k sub-experiments) has $N = N_1 \cdot N_2 \cdot N_3 \dots \cdot N_k$ possible outcomes.

Intuition: A Chipotle order.

Sub-exp. 1: Bowl or Burrito? , $N_1 = 2$

Sub-exp 2: White or Brown Rice? , $N_2 = 2$

Sub-exp 3: Black or Pinto Beans? , $N_3 = 2$

Sub-exp 4: Chicken, Steak, or Veggies? , $N_4 = 3$

$\Downarrow$
 $N = 2 \cdot 2 \cdot 2 \cdot 3 = \underline{24}$ different possible orders.

Ex: Selecting a Student Committee

Given $n=26$ students, we want to select three.

Sub-exp. 1: Choose 1st from 26 , $N_1 = 26$

Sub-exp. 2: Choose 2nd from 25 , $N_2 = 25$

Sub-exp. 3: Choose 3rd from 24 , $N_3 = 24$

$$\rightarrow N = 26 \cdot 25 \cdot 24 = \underline{15,600} \text{ possible outcomes?}$$

Note: Once we chose the 1st student, we are not allowed to choose them again. Hence, $N_2 = 26 - 1 = 25$.

This is called sampling without replacement.

An Observation: The resulting sample space of $N = 15,600$ possible outcomes does care about the order in which students were selected. I.e. assign a letter A-Z to each student, choosing A first, B second, and C third would be considered a different outcome from choosing C first, B second, and A third.

How many of the 15,600 outcomes have all of A, B, C?

$\rightarrow$ sub-exp. 1: Choose one (of 3) to be selected first

// 2: Choose one (of 2) to be selected second

// 3: Choose one (of 1) to be selected third

$\Rightarrow N = 3 \cdot 2 \cdot 1 = \underline{6}$ outcomes with the same three students.

Every unique set of 3 students will appear 6 times,

so we should have $N = \frac{15,600}{6} = \underline{2,600}$ outcomes?

To determine which of the two is correct, we need to answer: does the order matter?

Def. (Factorial): For any non-negative integer k , k Factorial, denoted $k!$, is defined as $k! = k \cdot (k-1) \cdot (k-2) \cdot \dots \cdot 2 \cdot 1$.

Def. (Permutation): For positive integers n, k , with $k \leq n$, the size k permutations of n , denoted $P(n, k)$, is:

$$P(n, k) = \frac{n!}{(n-k)!}$$

Def. (Combinations): For positive integers n, k , with $k \leq n$, the size k combinations of n , denoted $C(n, k)$ or $\binom{n}{k}$, is:

$$C(n, k) = \frac{n!}{(n-k)! k!}$$

Takeaway: When choosing k objects from n without replacement, we can count the number of possible outcomes as:

- $P(n, k)$ if the order matters
 - $C(n, k)$ if the order does not matter
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Counting Examples:

Ex. 1: From a 52-card deck, how many unique hands of 5 cards can be drawn?

Sampling? $\leadsto$ Without replacement

Order Matters? $\leadsto$ No

$$\Rightarrow \# \text{ of 5-card hands} = C(52, 5) = \frac{52!}{47! \cdot 5!} = \frac{52 \cdot 51 \cdot 50 \cdot 49 \cdot 48}{\underline{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}} = \underline{2,598,960}$$

Notice that the numerator corresponds to the product of 5 sub-experiments:

1: Choose 1 card from 52

2: Choose 1 card from 51

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5: Choose 1 card from 48

The denominator tells us how many of the 5-card hands contain the same set of 5 cards, but in different orders.

Ex. 2: How many 5-card hands have exactly 2 aces?

Sub-exp 1: Choose 2 aces from the 4 available

Sub-exp 2: Choose 3 cards from the 48 non-aces.

In both sub-exps, sampling is without replacement, and order doesn't matter.

$$\text{Outcomes for Sub-exp. 1: } C(4, 2) = \frac{4 \cdot 3 \cdot 2 \cdot 1}{(2 \cdot 1)(2 \cdot 1)} = \underline{6}$$

$$\text{Outcomes for Sub-exp. 2: } C(48, 3) = \frac{48!}{5! \cdot 45!} = \underline{17,296}$$

$$\leadsto \# \text{ Outcomes overall} = 6 \cdot 17,296 = \underline{103,776}$$

Back to Probability!

If every 5-card hand is equally likely, what is the probability of obtaining a hand with exactly 2 aces?

$$P(2 \text{ aces}) = \frac{\# \text{ hands w/ 2 aces}}{\# \text{ hands}} = \frac{103,776}{2,598,960} \approx \underline{0.04}$$
